DAFFODIL INTERNATIONAL UNIVERSITY



DEPARTMENT OF COMPUTING AND INFORMATION SYSTEM

LAB REPORT 2

Course Title : Object Oriented Programming

Course Code : CIS 216L

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OOP Assignment

Java Programming Basics

```
public class CommentsExample {
    public static void main(String[] args) {

        // Single-line comment
        System.out.println("Hello, World");

        /*
         * Multi-line comment
         */

        int x = 10;

        /**
         * Documentation comment
         */
        System.out.println("Value of x: " + x);
    }
}
```

Output

```
Hello, World
Value of x: 10
```

```
public class DataTypesDemo {
    public static void main(String[] args) {

        byte b = 100;
        short s = 30000;
        int i = 100000;
```

```
        long l = 10000000000L;

        float f = 3.14f;
        double d = 3.14159265359;
        char c = 'A';
        boolean flag = true;

        String str = "Hello, Java";
        int[] arr = {1, 2, 3, 4, 5};

        Integer wrapperInt = Integer.valueOf(50);
        StringBuilder sb = new StringBuilder("Java");

        System.out.println("byte: " + b);
        System.out.println("short: " + s);
        System.out.println("int: " + i);
        System.out.println("long: " + l);
        System.out.println("float: " + f);
        System.out.println("double: " + d);
        System.out.println("char: " + c);
        System.out.println("boolean: " + flag);
        System.out.println("String: " + str);

        System.out.print("Array: ");
        for (int num : arr) {
            System.out.print(num + " ");
        }

        System.out.println();
        System.out.println("Wrapper Integer: " + wrapperInt);
        System.out.println("StringBuilder: " + sb);
    }
}
```

Output

```
byte: 100
short: 30000
int: 100000
long: 10000000000
float: 3.14
double: 3.14159265359
char: A
boolean: true
String: Hello, Java
Array: 1 2 3 4 5
Wrapper Integer: 50
StringBuilder: Java
```

```
public class VariablesDemo {

    int instanceVar = 10;
    static String staticVar = "I am static";

    public void showVariables() {

        int localVar = 5;

        System.out.println("Instance Variable: " + instanceVar);
        System.out.println("Static Variable: " + staticVar);
        System.out.println("Local Variable: " + localVar);
    }

    public static void main(String[] args) {

        VariablesDemo obj1 = new VariablesDemo();
```

```
        obj1.showVariables();

        System.out.println(
            "Accessing Static Variable via class: "
            + VariablesDemo.staticVar
        );
    }
}
```

Output

```
Instance Variable: 10
Static Variable: I am static
Local Variable: 5
Accessing Static Variable via class: I am static
```

```
public class SimpleOperatorsDemo {
    public static void main(String[] args) {

        int a = 10, b = 3;

        System.out.println("a + b = " + (a + b));
        System.out.println("a - b = " + (a - b));

        System.out.println("a > b ? " + (a > b));

        boolean x = true, y = false;
        System.out.println("x && y = " + (x && y));

        a += 5;
        System.out.println("a after += 5 : " + a);

        int max = (a > b) ? a : b;
```

```
        System.out.println("Maximum = " + max);
    }
}
```

Output

```
a + b = 13
a - b = 7
a > b ? true
x && y = false
a after += 5 : 15
Maximum = 15
```

```
public class DecisionMakingDemo {
    public static void main(String[] args) {

        int number = 10;

        if (number > 0) {
            System.out.println("The number is positive.");
        }

        if (number % 2 == 0) {
            System.out.println("The number is even.");
        } else {
            System.out.println("The number is odd.");
        }

        if (number < 0) {
            System.out.println("The number is negative.");
        } else if (number == 0) {
            System.out.println("The number is zero.");
        } else {
            System.out.println("The number is positive.");
        }

        int day = 3;
```

```
        switch (day) {
            case 1:
                System.out.println("Monday");
                break;

            case 2:
                System.out.println("Tuesday");
                break;

            case 3:
                System.out.println("Wednesday");
                break;

            default:
                System.out.println("Other day");
        }
    }
}
```

Output

```
The number is positive.
The number is even.
The number is positive.
Wednesday
```

```
public class LoopsDemo {
    public static void main(String[] args) {

        // For loop
        for (int i = 1; i <= 5; i++) {
            System.out.println("i = " + i);
        }

        // While loop
```

```
        int j = 1;
        while (j <= 5) {
            System.out.println("j = " + j);
            j++;
        }

        // Do-While loop
        int k = 1;
        do {
            System.out.println("k = " + k);
            k++;
        } while (k <= 5);

        // Enhanced For Loop
        int[] numbers = {10, 20, 30, 40, 50};

        for (int num : numbers) {
            System.out.println("num = " + num);
        }
    }
}
```

```
public class LoopsDemo {
    public static void main(String[] args) {

        // 1. For loop
        System.out.println("For Loop:");

        for (int i = 1; i <= 5; i++) {
            System.out.println("i = " + i);
        }

        // 2. While loop
        System.out.println("\nWhile Loop:");

        int j = 1;

        while (j <= 5) {
            System.out.println("j = " + j);
            j++;
        }
```

```
        // 3. Do-While loop
        System.out.println("\nDo-While Loop:");

        int k = 1;

        do {
            System.out.println("k = " + k);
            k++;
        } while (k <= 5);

        // 4. Enhanced For Loop
        System.out.println("\nEnhanced For Loop:");

        int[] numbers = {10, 20, 30, 40, 50};

        for (int num : numbers) {
            System.out.println("num = " + num);
        }
    }
}
```

Output

```
For Loop:
i = 1
i = 2
i = 3
i = 4
i = 5

While Loop:
j = 1
j = 2
j = 3
j = 4
j = 5

Do-While Loop:
k = 1
k = 2
k = 3
k = 4
k = 5

Enhanced For Loop:
num = 10
num = 20
num = 30
num = 40
num = 50
```